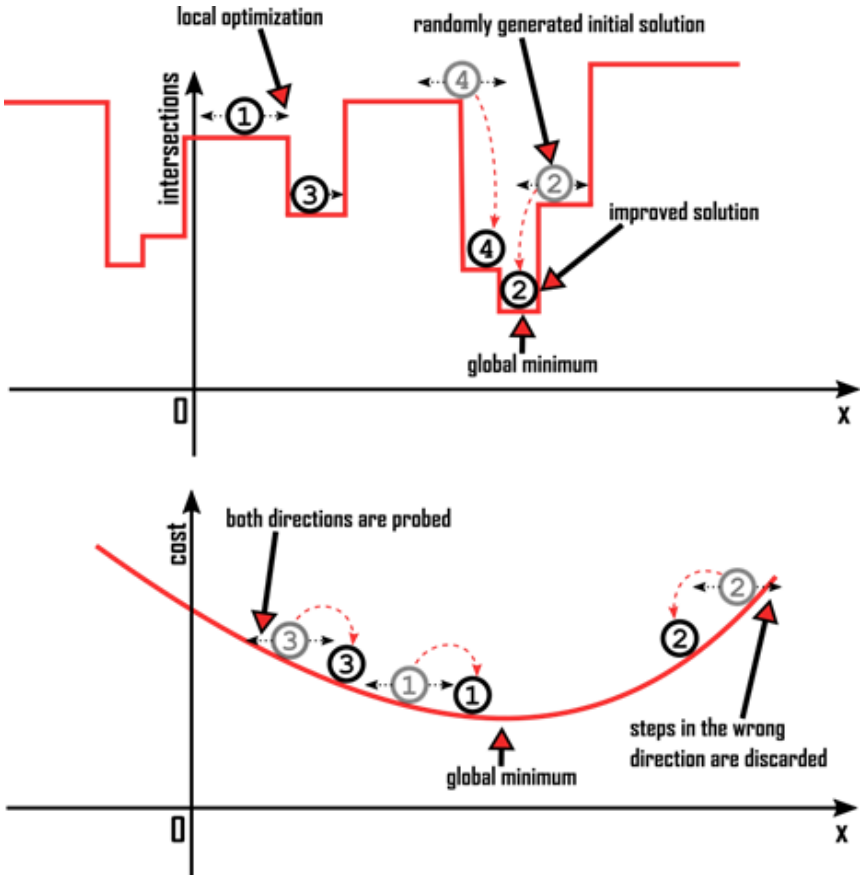


Figure 16.12 How a generic randomized heuristic with local optimization works, explained using the marble analogy. Notice that we need to try updated x in both directions, and check if we get an improvement.

Another thing that we must remark is that with this algorithm we have to try moving in both directions: by randomly exploring the area surrounding a solution, we are blindly poking similar solutions, but without any rationale. For instance, in the bottom example in figure 16.12, by looking at the marbles and the shape of the cost function we would know that, in order to get an improvement, we should increase the value of the only parameter for solutions 1 and 3, and decrease it for solution 2. Using *hill descent*, we would both try to increase and decrease it for all solutions, and just see what happens. For functions of more than a single parameter, we would either search the surrounding area